

SafeBatch AI — Furiosa A (Agentic Payment)

A Korean-language assistant that checks a user's transfer goal against sourced conditions and executes only a bounded TRON Nile testnet payment — declining otherwise.

- Track: Furiosa A · Model: Kiln **qwen3-32b** only · Chain: TRON **Nile testnet** (test funds) · Solo Builder
- Three demonstrated flows on 2026-09-29: **Normal** (1 USDT, confirmed) · **Adapt** (budget cut → 0.6 USDT, confirmed) · **Decline** (0 orders, 0 tx)
- Roles kept apart on every screen: **AI explains** · **code computes, verifies, blocks** · **human approves a fingerprint and signs**
- **Separate phone prototype** (same day): 1 Nile TRX transfer signed on an Android wallet, verified/broadcast once by the MacBook; its AI parsing and remote (LTE) access were verified **separately and partially** (p. 9).

Not claimed: exchange automation, "AI decided the payment", usability improvement, superiority over translation or general AI. Exchange guidance and the testnet payment are separate.

1 · Problem (one real user, honestly scoped)

- A Korean user moving USDT from a domestic exchange to their own overseas account faces English screens, many menus and conditions scattered across help pages (travel rule tiers, name matching, network choice, minimums).
- Evidence: the builder's own experience + one recorded test (2026-09-28) on a **synthetic** deposit screen: with literal translation the user could not tell *whose address* is shown or *what to do next*; our **first** guide screen failed to convey purpose at all.
- What we do **not** know: broader demand, whether the revised screen beats translation/general AI (not re-tested), multi-language needs.

Why an agent, not a FAQ

The same user later wants money to move. The useful unit is: **goal** → **checked conditions** → **a plan the user approves** → **execution that stops when conditions are not met**. That is exactly the Furiosa A pattern, so we built the payment part on a testnet with strict bounds.

2 · Product scope (what is in, what is out)

In

- Korean guide screen: free-text goal → Kiln structures it, asks the one missing input; code attaches condition cards with source URL, document date, checked time, confidence.
- First card after the goal: *the one action now / conditions to check first / why*, then one easy yes/no question to set the step (no step selector up front; the screen never reads the exchange). Added after the user test.
- Nile payment flows: read-only prepare → human-confirmed execute (TronLink signature on /sign) → resolve by the same txID.
- Rules: exact amount or adjust-within-budget, allow-list, absolute deadline, TRX total cap split into Energy limit + charged bandwidth.
- Phone prototype (separate code path guide/phone_*): chat request → phone wallet signs once → MacBook verifies and broadcasts → same-txID result. Scope of what was verified is on p. 9.

Out

- No exchange API, no automatic deposit/withdrawal, no screen OCR/overlay.
- No mainnet, no real funds; the test wallet sends to itself.
- GasFree / TRON C path exists in code but is deferred (no Nile API key).
- No energy/latency measurement claims (energy was consumed without a TRX burn on testnet; root cause unverified).
- No first-time wallet-connection test (all signatures used an already-connected TronLink).

3 · Who does what

Actor	Does	Cannot
Kiln qwen3-32b	Structure the goal; ask the missing input; for payments, read user rules + the code-computed candidate list and answer {"choice_index","reason"}; for declines, answer {"decision":"decline","reason"} after code has blocked.	Create candidates, change amount/address/fee/deadline, approve, sign, broadcast, unblock. Outputs outside the allowed JSON keys are rejected.
Code	Candidates from rules; amount adjustment only in adjust-within-budget mode; TRX cap split (281 B tx + 64 B result reserve = 345 B × 1000 sun); unsigned tx + independent protobuf decode (20 checks); confirm fingerprint over every shown condition + raw bytes + txID; signer recovery; pre-broadcast re-quote; one broadcast after ledger reservation; receipt reconciliation (4 cost bounds, Transfer log, solidity).	Sign, touch exchanges.
Human	Approve only if the fingerprint matches what is shown; sign once in TronLink; cancel.	—

Per-flow Kiln usage (prompt/completion tokens): normal 444/67 · adapt 813/61 · decline 319/53. Six Kiln calls in total across the event; logged in guide/logs/kiln_calls.jsonl without credentials.

4 · User test → what we changed

Found (2026-09-28, 1 user, synthetic screen, A then B)

- Translation (A): purpose understood; *"is this the sender's or receiver's address? there is no send button, I don't know what to do next."*
- Our first guide screen (B): *"just small facts listed — I can't tell what this screen is for."*
- Recorded honestly: order effect, question-explaining dialogue, no timing or superiority numbers, condition-text recall not scored as understanding.

단계: 잠정 · 3/5 도착 계정 확인 · AI 미사용(규칙만)

① 지금 할 행동 하나

이 화면의 주소(받는 주소)를 복사해 둔다. 이 화면에서는 그 밖에 할 일이 없습니다.

② 먼저 확인할 조건

- ⚠ 네트워크: **업비트 출금에서도 이 화면과 똑같이 TRC20(Tron)을 골라야 합니다. 다르면 코인이 사라질 수 있습니다.**
- 최소 수량: 화면에 적힌 최소 수량(예: 1 USDT) 미만은 입금 처리되지 않을 수 있습니다. 최소 수량을 맞췄다고 입금이 보장되지는 않습니다.
- 입금 방식: 받는 곳 화면에 적힌 입금 방식 조건(예: 스마트 컨트랙트 입금 지원 여부)을 그대로 확인합니다.
- 받는 계정: 현재 제품은 본인 계정 간 이동만 안내합니다(거래소 일반 규정과는 별개). '내 계정'이라고 하셨으므로 진술로 반영했고, 실제 명의 일치는 Binance 화면에서 직접 확인합니다.

③ 확인이 필요한 이유

받는 주소·네트워크가 틀리면 코인이 사라질 수 있고, 최소 수량·입금 방식·본인 계정 조건은 받는 곳 화면에만 적혀 있기 때문입니다.

한 가지만 여쭙니다 — 받는 곳(Binance)의 USDT 입금 화면을 열어 받는 주소를 확인했나요?

확인했어요

아직이요

단계는 사용자의 답으로만 정합니다. 이 화면은 거래소 화면을 읽거나 상태를 확인하지 않습니다. 답에 따라 아래 '지금 할 행동'이 바뀝니다. (지금은 3단계 기준으로 보여드립니다)

이 화면은 거래소 화면을 읽거나 상태를 확인하지 않습니다. 복사·출금·보내는 각 거래소 앱에서 직접 합니다.

Live first screen after the goal is typed (rules-only capture, 2026-09-29 03:0x): ① the one action now → ② conditions to check first → ③ why → one easy yes/no question decides the step; the screen never reads the exchange.

5 · Flow ① Normal — 1 USDT within all bounds

- Rules: exactly 1 USDT to own wallet, budget 1.0, TRX cap 5, deadline 06:55 KST.
- Code: single candidate; Energy fee_limit 4,655,000 + bandwidth max 345,000 = 5,000,000 sun.
- Kiln (00:56, call e851f510): "...TRX 수수료 한도 내에서 네트워크가 Nile 테스트넷인 유일한 옵션입니다." — explanation of the single candidate.
- Human: fingerprint ce741dfb...3cd3 approved; TronLink signature 01:01:53.
- Chain: tx a7c6b7340fde03af6b9ac00158d916b1ec90595ed00c0e0ac548eea7e770ca24, block **71361508**, SUCCESS, solidity-confirmed, Transfer 1 USDT, fee 345,000 sun (= predicted bandwidth charge).

(지갑 창의 계약-수량-fee limit 과 아래 표가 같고, 이 지문이 실행기 요약과 같을 때만 ② 서명)

fb2ef8d54b50212b50db820faf214e73eab76492a3732d882d878e5fcd

수량: 1.000000 USDT

과 TRX 수수료 한도는 별개입니다(수수료 상한 최대 5 TRX, USDT 아님).

	NILE 테스트넷 (chainId 3448148188) — 메인넷 TRC20 이 아닙니다. 거래소 출금란에 이 주소를 넣지 마세요.
정(EOA)	TEQh4L9pabnbW4UpHmxXveY31Q3FLsRHHz
	TEQh4L9pabnbW4UpHmxXveY31Q3FLsRHHz
크 계약	USDT · TXYZopYRdj2D9XRtbG411XZZ3kM5VkAeBf
(USDT)	1.000000 USDT (최소 단위 1000000)
출 상한	5.000000 TRX (이 거래 최대 5.000000 TRX = Energy 한도 + Bandwidth 최대)
한도(fee_limit)	4.655000 TRX · 예상 Energy 1.904500 TRX (14650 × 100 sun × 1.3)
th 최대	0.345000 TRX (과금 345 B = 전송 281 B + 결과 상한 64 B, × 1000 sun, 무료 대역폭 소진 가정) · 지금 예상 총 2.249500 TRX
만료	2026. 9. 29. 오전 6:55:33 (사용자 기한 2026. 9. 29. 오전 6:55:33)
	a7c6b7340fde03af6b9ac00158d916b1ec90595ed00c0e0ac

Ledger: intent DRAFTED → AWAITING_HUMAN → SIGNED → SUBMITTED → ACCEPTED → CONFIRMED; one broadcast; unknown responses are resolved by the same txID only.

Live: tx, receipt, solidity, Kiln call. **Synthetic:** flow tests with a fake node.

6 · Flow ② Adapt — budget cut 1.0 → 0.6 USDT

- Rules changed by the user: budget 0.6, minimum 0.5, mode **adjust within budget**. The screen shows **before** → **after** → **user's reason**.
- Code recomputes the candidate: **0.6 USDT** (adjustment is code, only in this mode; in exact mode it would be declined).
- Kiln (01:18, call a5f4bba8): "예산이 1.0 USDT에서 0.6 USDT로 줄어들었기 때문에, 후보는 최대 0.6 USDT를 보내는 것을 반영합니다."
- Honest note: that run stopped **before signing** (an executor fingerprint-revision bug, fixed with a regression test). The identical validated judgment was **carried** into the re-run (AI_CARRIED, 0 extra calls) — allowed only if the prior run ended pre-signature, the call exists in the log (ok/200/qwen3-32b), the preserved raw response re-parses to the same choice, and rules/plan/fingerprint are identical.
- Chain: tx 71a951bc63b73b3146f74d6b66b9b46b994be2f20a715ae57104acc0c83b4f88, block **71362119**, SUCCESS, solidity-confirmed, Transfer 0.6 USDT, fee 345,000 sun.

(지갑 창의 계약-수량-fee limit 과 아래 표가 같고, 이 지문이 실행기 요약과 같을 때만 ② 서명)

!13ee64f5962c889d5deff2c96cc62abd43e0e2cbd6461baaf6641114da

↳량: **0.600000 USDT** (요청 1.000000 → 이번 서명 수량 0.600000; 이전 수량은
₩에 없음)

₩ TRX 수수료 한도는 별개입니다(수수료 상한 최대 5 TRX, USDT 아님).

	NILE 테스트넷 (chainId 3448148188) — 메인넷 TRC20 이 아닙니다. 거래 소 출금란에 이 주소를 넣지 마세요.
정(EOA)	TEQh4L9pabnbW4UpHmxXveY31Q3FLsRHHz
	TEQh4L9pabnbW4UpHmxXveY31Q3FLsRHHz

/sign screen (replay render): before → after → user reason, AI_CARRIED with the original call id.

7 · Flow ③ Decline — rules cannot be met, nothing moves

- Rules: exactly 1 USDT, budget **0.6**, **no reduction allowed**.
- Code first: `candidate_plans` → 0 candidates → blocked: *"exact amount 1000000 + fee 0 exceeds budget 600000"*. No unsigned tx, no order, no intent, no approval. A pre-check run with **no AI call** already produced this block.
- Kiln (01:36, call `df785a78`): *"예산 0.6 USDT는 정확한 지급 금액 1 USDT보다 적어 결제가 불가능합니다."* — explanation only, no alternative amount generated; any other decision or extra key is ignored (tested).
- Result: **0 orders · 0 signatures · 0 broadcasts · 0 tx**; pending-order count and intent-ledger line count were the same before and after the run (no whole-directory hash was taken).

"Declined" is a first-class outcome: the record carries the code reason, the AI explanation, and the before/after ledger counts. **Live**: the Kiln call and the ledger snapshot. **Synthetic**: the decline-path unit tests (model output ignored when malformed; no call when a candidate exists).

8 · Safety properties held on Nile · Phone prototype stated as-is

Safety properties (Mac flows, real testnet)

- **Fingerprint-bound approval** over every shown condition + raw bytes + txID; any change → no execution (this stopped the first adapt run).
- **Signature checks**: raw bytes = unsigned original, txID recomputed, signer EOA = sender, one signature. **Single broadcast** after ledger reservation; unknown → same-txID lookup only.
- **Bandwidth charge model** (java-tron BandwidthProcessor) verified on 3 live receipts (345,000 sun as predicted). **Model containment**: shape-only validation; 224 + 105 synthetic tests.

Phone prototype — three separate facts

- ① **Phone-signed transfer (same Wi-Fi, 09:55 KST)**: 2 TRX, tx 178b51e6...7101c1, block 71372152, fee 0, solidity-confirmed. Parsing then was **rule-based, no model call**. Owner signed 4× over 3 orders; only the last was broadcast (others refused or never received).
- ② **Kiln parsing verified separately (13:51)**: 4/4 tool calls = 1 CLI format check + 3 requests sent **from the MacBook** to the same server route (normal · adapt · decline). 0 orders, 0 signatures; **not typed on the phone**. An earlier batch of 4 failed (reported).
- ③ **Remote access**: 127.0.0.1 server behind Tailscale Serve HTTPS (tailnet-only, Let's Encrypt). Verified from the MacBook (15:13) and **from the phone on LTE — existing-order lookup only** (16:29, screenshot + server log). Remote chat / signing: **unverified**.



Phone screen (owner capture, 09:40 KST, LAN): the order the wallet is asked to sign — receiver, amount, network, fee estimate and cap, expiry, txID, fingerprint, 19/19 server pre-checks. Paid-call gate: reservation before each model call; an unresolved one blocks the next; success only after schema/semantic/cost checks.

9 · Limits, pre-/during-event disclosure, next

Limits

- Self-transfer test wallet; no exchange integration; the guide's product link to the payment flows is an open question.
- One user, synthetic screen, no re-test → no usability-gain claim.
- Adapt flow's AI judgment carried from a pre-signature failed run (documented).
- First-time wallet-connect UX verified only with an already-connected wallet.
- Energy use (14,650/transfer) not burned from the wallet — cause unverified.
- Phone prototype: signing-time parsing was rule-based; AI parsing verified with MacBook-sent inputs; on LTE only an existing-order lookup was verified (no remote chat/signing).

Disclosure

- **Pre-event (2026-09-15):** CSV validation, budget/allow-list policy, approval-wait flow, 39 tests — no AI, no chain.
- **During event (09-28 16:30 → 09-29):** everything else — guide UI, Kiln client, Nile adapter, executor, /sign, ledgers, all three flows, and the phone prototype (09-29).

Next

- Re-test the revised screen against translation/general AI; connect the guide's goal to a payment plan only if that test supports it; first-connect UX check.